

Yes. Based on the **exact 5-slide structure you were given**, I would reorganize the PDF content like this. This keeps the important points from the document while making it suitable for an SIH-style PPT.

Slide 1 — Proposed Solution

DoSJE SmartMonitor (DSM)

AI-Powered Risk-Based Monitoring & Inspection Platform

Our Idea

DSM is an integrated digital platform designed to improve monitoring of NGOs/institutions funded under DoSJE schemes. It combines **CCTV, Edge AI, surprise inspections, GPS verification, blockchain-based evidence, analytics and beneficiary feedback** into one platform.

How It Addresses the Problem

- Replaces dependence on **monthly self-reported data** with continuous evidence-based monitoring.
- **AI people counting** cross-checks reported beneficiary/attendance numbers with actual presence.
- **Risk-based monitoring** focuses resources on high-risk NGOs while reducing unnecessary surveillance of compliant organizations.
- **Triple randomization** selects the NGO, inspector and inspection time, reducing predictable inspections and collusion.
- **Multi-source GPS verification** prevents fake inspection locations.
- **Blockchain hashing** protects inspection evidence from tampering.
- **Offline-first architecture + SMS fallback** enables operation in rural/low-connectivity areas.
- Beneficiaries can **verify enrollment, view grant information, submit anonymous grievances and rate services**.

Innovation & Uniqueness

1. **Risk-Based Tiering** — monitoring intensity changes according to fraud risk.
2. **Edge AI + Privacy-by-Design** — face blurring happens before footage leaves the premises.
3. **Triple Randomization** — NGO + inspector + inspection timing are randomized.
4. **Blockchain Evidence Chain** — creates tamper-evident inspection evidence.
5. **Multi-Source GPS Verification** — four independent location signals are cross-checked.
6. **Human-in-the-Loop AI** — officials verify alerts instead of blindly trusting automation.

7. Offline-First Design — designed specifically for India's connectivity challenges.

8. Vendor-Agnostic Architecture — open standards reduce vendor lock-in.

One-line USP

“From periodic, report-based inspection to continuous, risk-based and evidence-driven governance.”

Slide 2 — Technologies + Methodology

Technology Stack

Software

- **Mobile:** Flutter + Dart
- **Web Dashboard:** React + TypeScript + Vite
- **Backend:** Node.js + Fastify
- **AI/ML:** Python + FastAPI
- **Database:** PostgreSQL + PostGIS
- **AI:** YOLOv8 + LSTM/Prophet
- **CCTV:** RTSP + MediaMTX + FFmpeg
- **Video Conferencing:** Jitsi Meet
- **Authentication:** Keycloak
- **Storage:** MinIO
- **Monitoring:** Prometheus + Grafana
- **Infrastructure:** Docker/Kubernetes
- **Blockchain:** India Chain / Hyperledger

Hardware

- IP CCTV cameras
- NVR with Edge-AI capability
- GPS-enabled smartphones
- Bluetooth beacons for high-risk locations
- Local storage/HDD

- Internet/4G/5G where available

Implementation Methodology

NGO Registration

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Risk Scoring

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Tier 1 / Tier 2 / Tier 3 Monitoring

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CCTV + Edge AI Monitoring

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AI Detects Anomaly

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Human Official Verifies Alert

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Randomized Inspection Assigned

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GPS + Wi-Fi + Cell + Bluetooth Verification

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Digital Checklist + Photo/Video Evidence

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Blockchain Hashing

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DoSJE Dashboard & Analytics

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Beneficiary Verification + Grievance

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AI Retraining & Continuous Improvement

Prototype Status

The PDF reports an MVP containing:

- Flutter Android application
- Node.js + PostgreSQL backend
- 2 CP Plus IP-camera integration
- YOLOv8 people-counting model
- React real-time dashboard
- Blockchain hashing integration.

Best visual for this slide: Put the workflow above in the center and the technology stack as small icons around it.

Slide 3 — Feasibility, Challenges & Risk Mitigation

Feasibility Analysis

Area	Score	Assessment
Technical	8/10	Existing technologies and MVP available
Operational	5/10	Training/adoption required
Financial	7/10	PPP model improves sustainability
Legal	4/10	Privacy/legal authorization requires attention
Social	5/10	NGO and beneficiary acceptance required
Political	8/10	Strong alignment with transparency goals
Overall	6.3/10	Feasible with conditions

Key Challenges → Solutions

AI Accuracy

Challenge: False positives may increase inspector workload.

Solution: Human verification + monthly retraining + larger annotated datasets.

Privacy & Legal Risk

Challenge: CCTV surveillance may create constitutional/privacy concerns.

Solution: Face blurring, data minimization, 30-day automatic deletion, two-level unblurring approval and independent privacy oversight.

Rural Connectivity

Challenge: Some NGOs cannot support continuous streaming.

Solution: Offline-first mobile app + local NVR storage + scheduled upload + SMS fallback.

Inspector/NGO Resistance

Challenge: Learning curve and increased accountability may create resistance.

Solution: Training, district champions, incentives, co-creation workshops and tiered inspections.

Financial Risk

Challenge: Hardware costs, budget uncertainty and PPP failure.

Solution: PPP model, multi-vendor procurement, contingency fund and monthly cost monitoring.

Vendor Lock-in / Technology Obsolescence

Challenge: Dependence on a single vendor can make future upgrades expensive.

Solution: ONVIF, REST/GraphQL, PostgreSQL, ONNX and modular architecture.

Overall Residual Risk

4.2 / 10 — Medium-Low

Slide 4 — Potential Impact & Benefits

Impact on Target Audience



DoSJE Officials & Inspectors

- Reduced fieldwork and travel burden.
- Faster anomaly detection.
- Better evidence for decision-making.
- Improved job satisfaction.
- Digital tools for performance-based governance.



NGOs / Institutions

- Compliant NGOs receive **fewer unnecessary inspections**.
- Faster fund disbursement.
- Reduced paperwork.
- Incentive to improve compliance and service quality.
- Transparent performance measurement.



Beneficiaries

- Verify whether they are actually enrolled.
- See the grant amount allocated for their care.
- Anonymous grievance mechanism.

- Rate NGO services.
- Better food, education, healthcare and rehabilitation services through improved accountability.

Government & Taxpayers

- Reduced welfare leakage and fraud.
- Data-driven policy decisions.
- Better coordination between agencies.
- Reusable technology for other government schemes.
- Improved transparency and public trust.

Expected Benefits

Category	Expected Improvement
Welfare fraud	60% reduction
Inspection time	37% faster
Beneficiary satisfaction	60% → 80%
NGO compliance	65% → 85%
Inspector satisfaction	65% → 82%
Fund disbursement	30 → 7 days
Paper consumption	83% reduction
CO ₂ emissions	85% reduction
Public trust	45% → 65% projected

Long-Term Impact

DSM's modular components could potentially be adapted for **MGNREGA attendance, PDS monitoring and government-school attendance**, creating a reusable digital-governance framework rather than a one-time solution.

Slide 5 — References & Research Work

References / Research Sources

Government & Legal References

- **Supreme Court of India — K.S. Puttaswamy v. Union of India (2017)**
Right to Privacy and proportionality principles.

- **Ministry of Electronics & IT — Digital Personal Data Protection framework**
- **NITI Aayog — Responsible AI for Social Empowerment (RAISE 2025)**
- **Right to Information Act, 2005**
- **DoSJE schemes and monitoring systems referenced in the research**

International Research

- **World Bank — Leveraging Technology for Social Protection: Global Lessons (2025)**
- **UNDP Asia-Pacific — Digital Governance for SDGs: Case Studies from Asia (2026)**
- **Transparency International — Government Defence Anti-Corruption Index (2025)**

The research identifies alignment with:

- **SDG 1 — No Poverty**
- **SDG 10 — Reduced Inequalities**
- **SDG 16 — Peace, Justice & Strong Institutions**

Open-Source Technology References

- **MediaMTX** — CCTV streaming
- **Ultralytics YOLOv8** — object detection / people counting
- **Jitsi Meet** — video conferencing
- **Keycloak** — identity and access management
- **MinIO** — object storage
- **PostgreSQL / PostGIS** — database and geospatial data

Research & Prototype Validation

Pilot: 5 local NGOs, 4-week trial

Inspector survey: 15 respondents

NGO feedback: 5 NGO focus groups

Prototype: Flutter + Node.js + PostgreSQL + CCTV + YOLOv8 + React + blockchain.

One Important Recommendation

For the actual PPT, **don't copy these paragraphs directly**. Use them as your content/script.

I would structure the visual presentation like this:

Slide 1:

PROBLEM → DSM → 8 INNOVATIONS

Slide 2:

TECH STACK + COMPLETE WORKFLOW DIAGRAM

Slide 3:

FEASIBILITY SCORECARD + CHALLENGE → MITIGATION

Slide 4:

4 TARGET GROUPS + BIG IMPACT NUMBERS

Slide 5:

LEGAL REFERENCES + RESEARCH + OPEN-SOURCE REFERENCES